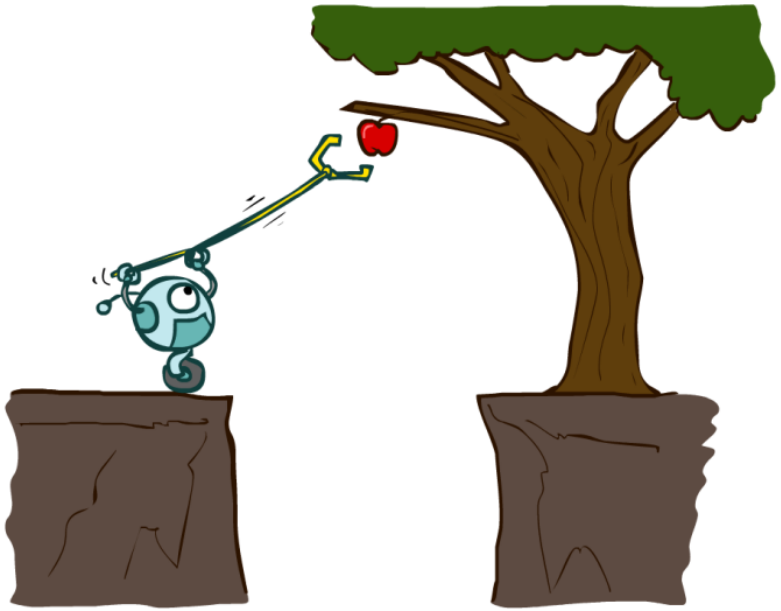
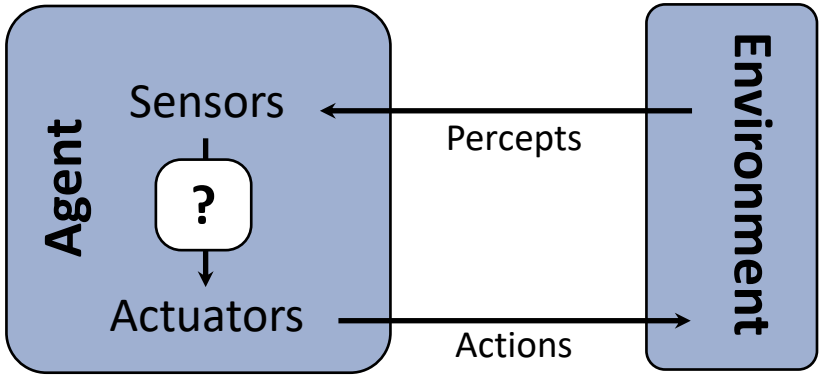
Designing Rational Agents

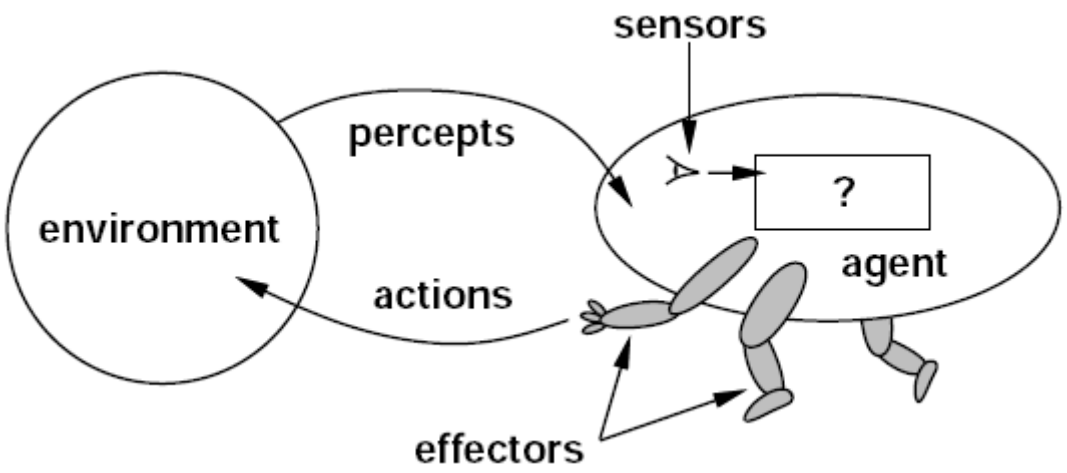
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions
- **This course** is about:
  - General AI techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique





What is an Agent?

- *"Intelligent agents continuously perform three functions: **perception** of dynamic conditions in the environment; **action** to affect conditions in the environment; and **reasoning** to interpret perceptions, solve problems, draw inferences, and determine actions."*



Intelligent Agents

- An intelligent agent may learn from the environment to achieve their goals.
- Following are the main four rules for an AI agent:
  - **Rule 1:** An AI agent must have the ability to perceive the environment.
  - **Rule 2:** The observation must be used to make decisions.
  - **Rule 3:** Decision should result in an action.
  - **Rule 4:** The action taken by an AI agent must be a rational action.

Example of Agents

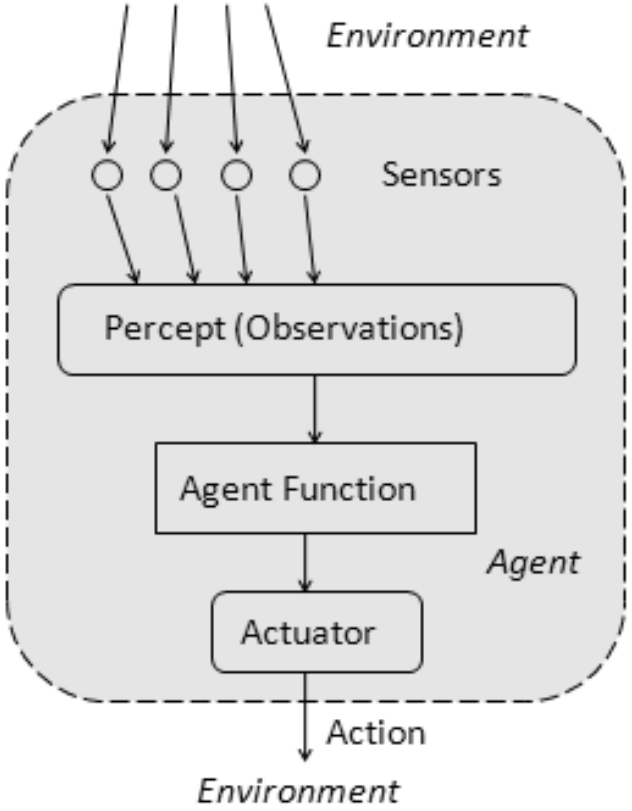
- Human Agent
  - eyes, ears, skin, taste buds, etc. for sensors
  - hands, fingers, legs, mouth, etc. for actuators
- Robot
  - camera, infrared, bumper, etc. for sensors
  - grippers, wheels, lights, speakers, etc. for actuators
- Software Agent
  - functions as sensors
    - information provided as input to functions in the form of encoded bit strings or symbols
  - functions as actuators
    - results deliver the output

Rational Agent

- **Rational Agent**:
  - For each possible percept sequence, a rational agent should select an action that is **expected to maximize its performance measure**, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.
- **Rationality** is distinct from **omniscience** (all-knowing with infinite knowledge)
  - **Actual (perfection) vs. Expected**
  - **Best vs. Optimal**

Agent Function and Program

- **Behavior of Agent** – It is the action that agent performs after any given sequence of percepts
- Agent's behavior is *mathematically* described by
  - **Agent function**
  - A function mapping any given percept sequence to an action
- *Practically* it is described by
  - An **agent program**
  - The real implementation
- Problems:
  - what is " the right thing"
  - how do you measure the "best outcome"



Specifying the Task Environment

Performance of Agents

Performance of Agents

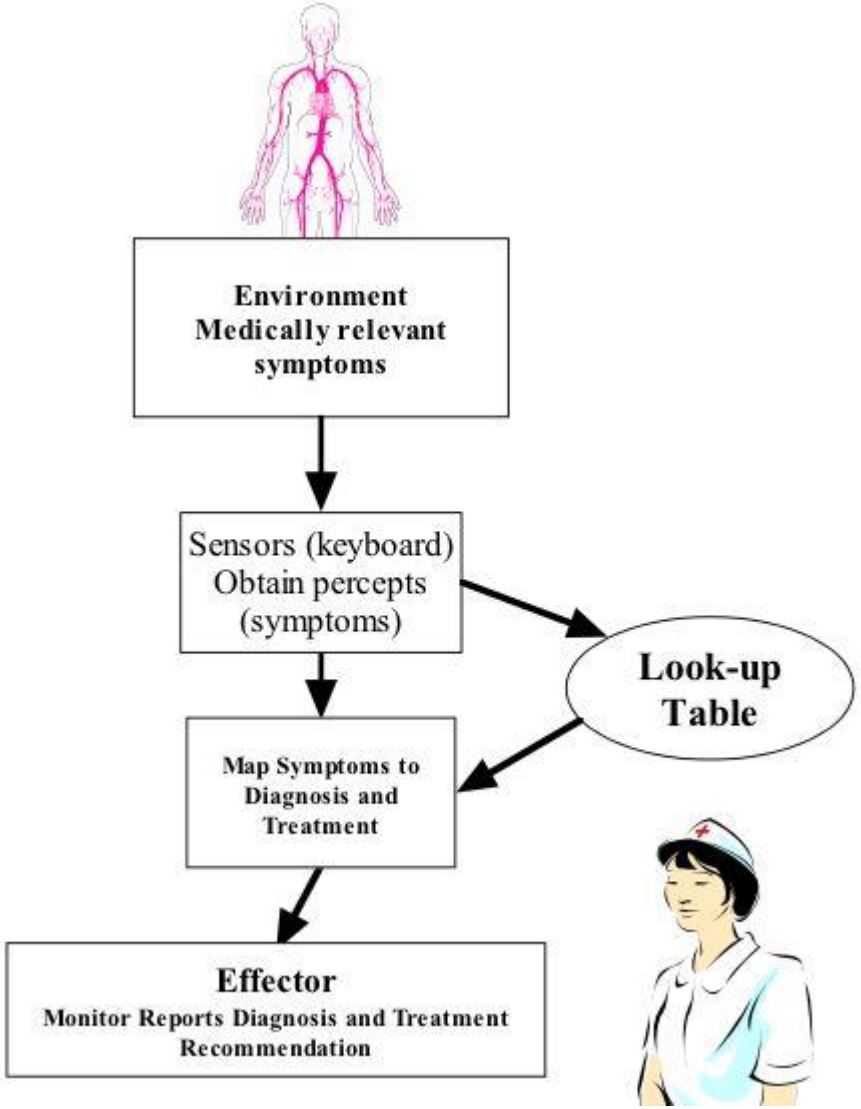
- Behaviour and performance of AI in terms of agent program.
  - Perception to Action Mapping
  - Ideal mapping: specifies which actions an agent ought to take at any point in time.
- Performance measure: a subjective measure to characterize how successful an agent is (e.g., speed, power, accuracy, etc.)
- **PEAS:**
  - **Performance measure,**
  - **Environment,**
  - **Actuators,**
  - **Sensors**

PEAS: Automated Taxi

- Performance measure
  - Correct Destination, Fuel, time, cost, violations, safety, comfort, profit
- Environment
  - rural, urban, other drivers, customers
- Actuators
  - Steering, brake, fuel, display/speaker
- Sensors
  - Camera, radar, accelerometer, engine sensors, microphone

PEAS: Medical diagnosis system

- Performance measure:
  - Healthy patient, minimize costs, lawsuits
- Environment:
  - Patient, hospital, staff
- Actuators:
  - Screen display (questions, tests, diagnoses, treatments, referrals)
- Sensors:
  - Keyboard (entry of symptoms, findings, patient's answers)



PEAS: Part-picking robot

- Performance measure:
  - Percentage of parts in correct bins
- Environment:
  - Conveyor belt with parts, bins
- Actuators:
  - Jointed arm and hand
- Sensors:
  - Camera, joint angle sensors

PEAS: Interactive English tutor

- Performance measure:
  - Maximize student's score on test
- Environment:
  - Set of students
- Actuators:
  - Screen display (exercises, suggestions, corrections)
- Sensors:
  - Keyboard

Environment Types

Properties of environments

vs.

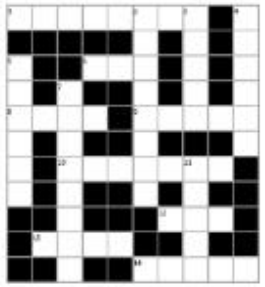
- **Fully observable** vs. **partially observable**
- **Or Accessible vs. inaccessible**
  - Can the agent observe the complete state of the environment?
  - If an agent's sensors gives it access to the complete state of the environment, then we say that environment is fully observable to the agent.
  - A fully observable environment is convenient because the agent need not maintain any internal state to keep track of the world.
  - ***Is the current state of the world fully known at each step?***

Properties of environments

- **Deterministic** vs. **nondeterministic (Stochastic).**
  - Is there uncertainty in how the world works?
  - If the next state of the environment is completely determined by the current state and the actions selected by the agents, then we say the environment is deterministic.
  - If the environment is inaccessible, then it may *appear* to be nondeterministic (bunch of uncertainties).

  - **Taxi driving** is clearly stochastic in this sense, because one can never predict the behavior of traffic exactly.
    - having a random probability distribution or pattern that may be analysed statistically but may not be predicted precisely.

Environment Types



vs.

- **Episodic** vs. **Sequential**:
  - Episodic: next episode doesn’t depend on previous actions.
  - The agent's experience is divided into atomic "episodes"
  - each episode consists of the agent perceiving and then performing a single action, and
  - the choice of action in each episode depends only on the episode itself.
  - the next episode does not depend on the actions taken in previous episodes
  - For example, an agent that has to spot defective parts on an assembly line (EPISODIC)
  - Chess and Taxi driver (SEQUENTIAL).

Environment Types

- **Static** vs. **Dynamic**:
  - If the environment can change while an agent is agent is performing action, then we say the environment is dynamic for that agent; otherwise, it is static.
  - Dynamic environments are continuously asking the agent what it wants to do
  - Static environments are easy to deal with, because the agent does not keep on looking at the environment while it is deciding on an action.
  - The environment is **semi-dynamic** if the environment itself does not change with the passage of time but the agent's performance score does.

Environment Types

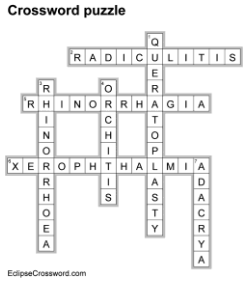
- **Discrete** vs. **continuous**:
  - Is there a finite (or countable) number of possible environment states?
  - The environment is discrete if the number of actions and possible states of the environment is finite otherwise it is continuous.
  - If there are a limited number of distinct, clearly defined percepts and actions, we say that the environment is discrete.
    - Chess, since there are a fixed number of possible moves on each turn.
    - Taxi driving is continuous.

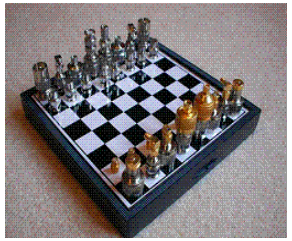
Types of Environment

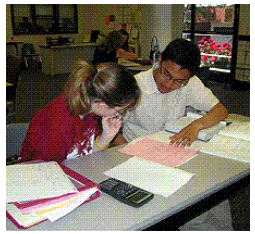
- **Single Agent** vs. **Multi Agent**
  - Is the agent the only thing acting in the world?
  - In the single agent environment there is only one agent
    - A computer software playing crossword puzzle
  - In multiagent systems, there are more than one active agents
    - Video games

vs.

Environment Types Example







	Crossword puzzle	Chess	Taxi driving	Part-picking robot	English tutor
Observable	Yes	Yes	No	No	No
Deterministic	Yes	Strategic	No	No	No
Episodic	No	No	No	Yes	No
Static	Yes	Static	No	No	No
Discrete	Yes	Yes	No	No	Yes
Agents	Single	Multi	Multi	Single	Multi